Discovering the Dilogarithm

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Abstract

This research project aims to deepen the understanding of the dilogarithm function by building on Don Zagier's article, "The Dilogarithm " [3] using methods from complex analysis [2] and algebra. The primary objective is to discuss and provide rigorous proofs for arguments or claims that were not fully addressed in Zagier's article. We begin by exploring the analytic continuation of polylogarithms. Then we will discuss various properties of the dilogarithm. Following this analysis, with our knowledge of the dilogarithm, we will discuss the Bloch-Wigner function, and examine its advantages. Finally, we will introduce the five-term relation and show how it can be used to prove properties of the dilograithim without the need of any complex analysis.

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1 Introduction

In this paper we will study the dilograrithim, initially defined over 250 years ago as the the following power series:

$$\sum_{n=0}^{\infty} \frac{z^n}{n^2} \qquad \text{for } \ |z| < 1$$

At its conception, the function went relatively unnoticed. However, with its appearance in newer fields of mathematics, such as K-theory, the dilograthim's relevance has gained momentum motivating this project. The objective of this project will be to explore and uncover the fascinating phenomena of the dilogarithm discussed in Don Zagier's article,"The Dilogarithm "[3] and prove some propositions that where omitted form the paper. In this paper, we embark on a journey which will consist of four parts. In Part One we will start off this project by inspecting a more generalized version of the dilogarithm, the polylogarithm. In this chapter we will define and prove the analytic extension of the polylogarithm. Armed with this new definition of a polylograithim, Part Two will define and prove interesting properties of the the dilograithim, using complex analysis where we come across the legendary Five-term relation. Part Three introduces the Bloch-Wigner function that has the same properties as the dilogarithm but is continuous on a greater domain and has simpler equalities. Finally, Part Four will analyze how the Five-term relation allows us to prove properties of Bloch-Wigner functions without the need for complex analysis tools.

2 Part One: Polylogarithms

Definition 1. *A polylogarithm is a power series defined as follows, let $k \in \mathbb{Z}$, $z \in \mathbb{C}$:*

$$Li_k(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^k}$$

When considering polylogarithms in $\mathbb{C}$, through the ratio test, it can be proven that $Li_k(z)$ is analytic for a given set of z as shown below:

$$\begin{aligned} \lim_{n\to\infty} \left| \frac{\frac{z^{n+1}}{(n+1)^k}}{\frac{z^n}{n^k}} \right| &= \lim_{n\to\infty} |\frac{z \cdot n^k}{(n+1)^k}| \\ &= |z| \end{aligned}$$

Hence, by the ratio test [2] for $|z| < 1$ $Li_k(z)$ is analytic. However, this function can be analytically extended. For example, notice that when $k = 1$:

$$\begin{aligned} \log(1-z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(-z)^n}{n} &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(-z)^n}{n} \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{2n-1}(z)^n}{n} \\ &= \sum_{n=1}^{\infty} \frac{-(z)^n}{n} && \text{As } 2n-1 \text{ is always odd} \\ &= -Li_1(z) && \text{By definition of polylogarithms} \end{aligned}$$

Hence, $-\log(1-z) = Li_1(z)$ as logs are analytic on $\mathbb{C}\backslash[0,\infty)$ by the principle of analytic continuation, $Li_1(z)$ can be extended to $\mathbb{C}\backslash[1,\infty)$. Hence we can redefine $Li_1(z)$ as follows:

$$Li_1 : \mathbb{C}\backslash[1,\infty) \to \mathbb{C} \quad \textit{such that} \quad Li_1(z) = \begin{cases} \sum_{n=1}^{\infty} \frac{z^n}{n} & \text{if } |z| < 1 \\ -\log(1-z) & \text{if } |z| \geq 1 \end{cases}$$

Which poses the question if an extension can be applied to all $Li_k(z)$, $k \geq 1$? In fact, we will demonstrate that it can be. Naturally, we would need to find what function would allow for this extension. When looking at the power series of $Li_k(z)$ for any $k > 1$, no familiar power series as in the case in $Li_1(z)$ seems to appear. Clearly more needs to be known about $Li_k(z)$ $(k > 1)$. Here, the derivative property of polylogarithm s, this proposition will shed light on this question. This intriguing property produces an elegant relationship between polylogrithims of different parameters .

Proposition 1 (Derivative property of polylogarithms)**.**

$$\frac{d}{dz} Li_k(z) = \frac{1}{z} Li_{k-1}(z)$$

Proof.

$$\begin{aligned} \frac{d}{dz} Li_k(z) &= \frac{d}{dz} \sum_{n=1}^{\infty} \frac{z^n}{n^k} \\ &= \sum_{n=1}^{\infty} \frac{nz^{n-1}}{n^k} \\ &= \sum_{n=1}^{\infty} \frac{z^{n-1}}{n^{k-1}} \\ &= z^{-1} \sum_{n=1}^{\infty} \frac{z^n}{n^{k-1}} \\ &= z^{-1} Li_{k-1}(z) \end{aligned}$$

□

Now geared with this property a second relationship between polylogarithm s of differing parameters follows, The integral Definition Of Polylogarithm s.

Proposition 2 (Integral property of polylogarithm s)**.**

$$\int_0^z \frac{1}{t} Li_k(t) dt = Li_{k+1}(z)$$

Proof.

$$\begin{aligned} \int_0^z \frac{1}{t} Li_k(t) dt &= \int_0^z \frac{d}{dt} Li_k(t) dt && \text{By Proposition 1} \\ &= Li_k(z) && \text{By Fundamental theorem of calculus} \end{aligned}$$

□

This proposition will play a crucial role in finding the analytic continuation that we were searching for.

Proposition 3 (The analytic extension of $Li_k(z)$). *For any* $k \in \mathbb{N}^+$,

$$F_k : \mathbb{C}\backslash[1,\infty) \to \mathbb{C} \quad \textit{such that} \quad F_k(z) = \begin{cases} -\log(1-z) & \textit{if } k = 1 \\ \int_0^z \frac{F_{n-1}(z)}{z} dz & \textit{if } k > 1 \end{cases}$$

$$Li_k : \mathbb{C}\backslash[1,\infty) \to \mathbb{C} \quad \textit{such that} \quad Li_k(z) = \begin{cases} \sum_{n=1}^{\infty} \frac{z^n}{n^k} & \textit{if } |z| < 1 \\ F_k(z) & \textit{if } |z| \geq 1 \end{cases}$$

Where the proposition will be proven by induction on the parameter of $Li_k(z)$.

Proof. The base Case of $n = 1$ is proven previously.

Now suppose for inductive hypothesis that: for a $n \in \mathbb{N}^+$

$$Li_n : \mathbb{C}\backslash[1,\infty) \to \infty \quad \textit{such that} \quad Li_n(z) = \begin{cases} \sum_{i=1}^{\infty} \frac{z^i}{i^n} & \text{if } |z| < 1 \\ F_n(z) & \text{if } |z| \geq 1 \end{cases}$$

Now, given by Integral Properties of Polylogarithm s,

$$\int_0^z \frac{Li_n(t)}{t} dt = Li_{n+1}(z)$$

Thus,

$$Li_{n+1}(z) = \begin{cases} \sum_{i=1}^{\infty} \frac{z^i}{i^{n+1}} & \text{if } |z| < 1 \\ \int_0^z \frac{F_n(t)}{t} dt & \text{if } |z| \geq 1 \end{cases} \quad = \begin{cases} \sum_{i=1}^{\infty} \frac{z^i}{i^{n+1}} & \text{if } |z| < 1 \\ F_{n+1}(z) & \text{if } |z| \geq 1 \end{cases}$$

□

3 Part Two: The Dilogarithm

The dilogarithm also known as $Li_2(z)$ has several interesting properties. The first property we claim that the dilogarithm can be continuously extended to $\mathbb{C}\backslash(1,\infty)$ (however not analytically), by stating $Li_2(1) = \frac{\pi^2}{6}$. We will shall prove this by observing the behaviour of Li_2(z) as z approaches 1. We start by considering the specifically when a sequence z_n approaches 1 from inside the unit disk (in the complex plane) .

For sufficiently large N for all $n > N$ there exists a $\delta > 0$ such that $1 - \delta < |z_n| < 1$ and thus:

$$\frac{\pi^2}{6} - \sum_{k=1}^{\infty} \frac{|z_n|^k}{k^2} =$$
$$= \sum_{k=1}^{\infty} \frac{1 - |z_n|^k}{k^2}$$
$$\leq \sum_{k=1}^{\infty} \frac{|1 - z_n^k|}{k^2} \qquad \text{By revese triangle inequality}$$
$$= \sum_{k=1}^{\infty} \frac{1 - 1 - P(\delta)}{k^2}$$

where $P(\delta)$ is a polynomial of order n and $P(\delta)$ has a zero of order 1

$$= \delta \sum_{n=1}^{\infty} \frac{-P^*(\delta) + c}{n^2}$$

Where c is a constant such that $P^*(\delta) + c = \dfrac{P(\delta)}{\delta}$

$$= \delta \sum_{n=1}^{\infty} \frac{-\delta - P^*(\delta)}{n^2} + \delta \sum_{n=1}^{\infty} \frac{c}{n^2}$$

As z_n approaches 1, δ goes to zero and therefore the expression above will approach zero as well. So it can be concluded, as z_n approaches 1 (for all $n \in \mathbb{N}$ $|z_n| < 1$) $\sum_{k=1}^{\infty} \frac{|z_n|^k}{k^2}$ approaches $\frac{\pi^2}{6}$.

When considering a sequence approaching 1 from outside the unit disk (excluding the $(1, \infty)$ branch cut). As $Li_2(z)$ is proven to be continuous on $\mathbb{C}\backslash[1, \infty)$; thus, the sequence $Li_2(z_n)$ will approach $\frac{\pi^2}{6}$ as well. Therefore, $Li_2(z)$ can be continuously extended to $\mathbb{C}\backslash(1, \infty)$, ([3]).

Proposition 4 (Property: Functional identity)**.**

$$Li_2(z) + Li_2(1 - z) = \frac{\pi^2}{6} - \log(z)\log(1 - z)$$

Proof. Firstly, consider $f(z) = \Big(Li_2(z) + Li_2(1 - z)) - (-\log(z)\log(1 - z)\Big)$, where when taking the derivative, the following expression can be deduced:

$$\frac{df(z)}{dz} =$$
$$= \frac{d}{dz}\Big(Li_2(z) + Li_2(1 - z)\Big) - \frac{d}{dz}\Big(-\log(z)\log(1 - z)\Big)$$
$$= \frac{d}{dz}Li_2(z) + \frac{d}{dz}Li_2(1 - z) - \frac{d}{dz}\Big(-\log(z)\log(1 - z)\Big)$$
$$= \frac{1}{z}Li_1(z) - \frac{1}{1 - z}Li_1(1 - z) - (\frac{\log(z)}{1 - z} - \frac{\log(1 - z)}{z}) \qquad \text{By derivative property}$$
$$= -\frac{1}{z}\log(1 - z) + \frac{1}{1 - z}\log(1 - (1 - z)) - (\frac{\log(z)}{1 - z} - \frac{\log(1 - z)}{z}) \qquad \text{As } Li_1(z) = -\log(1 - z)$$
$$= 0$$

As $f^{'}(z) = 0$, this implies that $\Big(Li_2(z) + Li_2(1-z)\Big)$ and $-\log(z)\log(1-z)$ differ by a constant, $c \in \mathbb{C}$. Where c can be deduced by setting $z = 1$:

$$c = Li_2(1) + Li_2(1-1) - (-\log(1)\log(1-1))$$

$$\Leftrightarrow c = \sum_{n=1}^{\infty} \frac{1}{n^2} + 0 - (-\lim_{z\to 1} \log(1-z)\log(z))$$

$$\Leftrightarrow c = Li_2(1) = \frac{\pi^2}{6}$$

Hence, it can be concluded that $(Li_2(z) + Li_2(1-z)) - (-\log(z)\log(1-z)) = \frac{\pi^2}{6}$. Which can be rearranged to provide the desired equality:

$$(Li_2(z) + Li_2(1-z)) = \frac{\pi^2}{6} - \log(z)\log(1-z)$$

□

Proposition 5.

$$Li_2\left(\frac{1}{z}\right) = -Li_2(z) - \frac{\pi^2}{6} - \frac{1}{2}\log^2(1-z)$$

Proof. Consider $\frac{d}{dz}(Li_2(\frac{1}{z})) - \frac{d}{dz}(-Li_2(z) - \frac{1}{2}\log^2(-z))$, where the following can be deduced:

$$\begin{aligned}
&\frac{d}{dz}Li_2\left(\frac{1}{z}\right) - \frac{d}{dz}\left(-Li_2(z) - \frac{1}{2}\log^2(-z)\right) = \\
&= \frac{d}{dz}\left(\frac{1}{z}\right)\frac{d}{dz}\left(Li_2\left(\frac{1}{z}\right)\right) + \frac{d}{dz}\big(Li_2(z)\big) + \frac{1}{2}\left(\frac{d}{dz}\big(\log^2(-z)\big)\right) && \text{By chain rule} \\
&= -\frac{1}{z^2}\left(\frac{Li_1(\frac{1}{z})}{\frac{1}{z}}\right) + \frac{d}{dz}Li_2(z) + \frac{1}{2}\left(\frac{d}{dz}\big(\log^2(-z)\big)\right) \\
&= -\left(\frac{Li_1(\frac{1}{z})}{z}\right) + \frac{d}{dz}Li_2(z) + \frac{1}{2}\left(-\frac{2\log(-z)}{-z}\right) \\
&= -\left(\frac{Li_1(\frac{1}{z})}{z}\right) + \frac{Li_1(z)}{z} + \frac{1}{2}\left(-\frac{2\log(-z)}{-z}\right) \\
&= \frac{\log(1-\frac{1}{z})}{z} + \frac{-\log(1-z)}{z} + \frac{\log(-z)}{z} \\
&= \frac{\log(\frac{1-z}{-z})}{z} + \frac{-\log(1-z)}{z} + \frac{\log(-z)}{z} \\
&= 0
\end{aligned}$$

Therefore, it has been deduced that $(Li_2(\frac{1}{z}))$ and $\big(-Li_2(z) - \frac{1}{2}\log^2(-z)\big)$ differ by a constant c. Thus, by setting $z = 1$, c can be computed as follows:

$$c = Li_2(\frac{1}{1}) - (-Li_2(1) - \frac{1}{2}\log^2(-1)) \Leftrightarrow c = \frac{\pi^2}{3} + \frac{1}{2}(\log(1) + \pi i)^2$$

$$\Leftrightarrow c = \frac{\pi^2}{3} - \frac{\pi^2}{2}$$

$$\Leftrightarrow c = -\frac{\pi^2}{6}$$

‘ This implies that:

$$-\frac{\pi^2}{6} = (Li_2(\frac{1}{z})) - (-Li_2(z) - \frac{1}{2}\log^2(-z)) \Leftrightarrow Li_2\big(\frac{1}{z}\big) = -Li_2(z) - \frac{\pi^2}{6} - \frac{1}{2}\log^2(1-z)$$

□

Proposition 6 (The duplication formula)**.** *For* $z \in \mathbb{C}\backslash\Big((-\infty,1)\cup(1,\infty)\Big)$

$$2(Li_2(z)+Li_2(-z)) = Li_2(z^2)$$

Proof. Suppose that $|z|<1$

$$\begin{aligned}
2(Li_2(z)+Li_2(-z)) &= 2\sum_{n=1}^{\infty}(\frac{z^n+(-z)^n}{n^2}) \\
&= 2\sum_{n=1}^{\infty}\frac{2(z)^{2n}}{(2n)^2} && \text{As for odd } n,\ z^n+(-z)^n = z^n-(z^n)=0 \\
&= \sum_{n=1}^{\infty}\frac{z^{2n}}{n^2} \\
&= Li_2(z^2)
\end{aligned}$$

Then by analytic continuation as $2(Li_2(z)+Li(-z))$ and $Li_2(z)$ agree on an open subset of $\mathbb{C}\backslash\Big([-\infty,1)\cup[1,\infty)\Big)$, $\{z\in\mathbb{C} \quad | \quad |z|<1\}$; thus, they must be equal on $\mathbb{C}\backslash\Big([-\infty,1)\cup[1,\infty)\Big)$. Finally, at $z=-1,1$ the expression the same, value

$$2(Li_2(1)+Li_2(-1)) = 2(\frac{\pi^2}{6}+Li_2(-1))$$

Where as $Li_2(z)$ is continuous on $\mathbb{C}\backslash(1,\infty)$, and as $\sum_{i=0}^{\infty}\frac{(-1)^i}{i^2} = -\frac{\pi^2}{12}$ thus by properties of continuity $Li_2(z) = -\frac{\pi^2}{12}$. Therefore:

$$= 2(\frac{\pi^2}{6}-\frac{\pi^2}{12}) = \frac{\pi^2}{6} = Li_2(1^2)$$

□

Proposition 7 (5-term relation)**.**

$$Li_2(x)+Li_2(y)+Li_2(\frac{1-x}{1-xy})+Li_2(1-xy)+Li_2(\frac{1-y}{1-xy}) = \frac{\pi^2}{6}-\log(x)\log(1-x)-\log(y)\log(1-y)+\log(\frac{1-x}{1-xy})\log(\frac{1-y}{1-xy})$$

The proof here [3] follows similar idea as the proof for the functional identity, except with a slight twist. Here, we take the x and y partials of of $Li_2(x)+Li_2(y)+Li_2(\frac{1-x}{1-xy})+Li_2(1-xy)+Li_2(\frac{1-y}{1-xy})$ and $\frac{\pi^2}{6}-\log(x)\log(1-x)-\log(y)\log(1-y)+\log(\frac{1-x}{1-xy})\log(\frac{1-y}{1-xy})$ and where they will both agree which implies that the two equations differ by a constant and that constant can be evaluated by setting $x=y$ and previous propositions.

4 Part Three: Bloch-Wigner Equations

The Bloch-Wigner Function[3] is defined as $D:\mathbb{C}\to\mathbb{R}$ such that $D(z) = \Im(Li_2(z)+iarg(1-z)\log|z|)$. This function displays some interesting properties, one of these powerful properties is that D is continuous on $\mathbb{C}\backslash\{0,1\}$. To motivate why such a phenomena occurs, we will first discuss as a simpler case. We will examine how $\log(z) = log|z|+i\arg_{(-\pi,\pi)}(z)$ which is continuous on $\mathbb{C}\backslash(-\infty,0)$ can be modified to become analytic over a greater domain. The reason why log isn't continuous on $(-\infty,0)$ is because there is a jump discontinuity of $2\pi i$ at $z_0\in(-\infty,0)$

which we will now justify. We start by considering the integral definition of log: define, $\gamma_1, \gamma_2 : [0,1] \to \mathbb{C}\backslash(-\infty, 0)$ (that are smooth on their domain) such that $\gamma_1(z)[0,1]$ is a curve $\gamma_1(0) = 1$ and $\gamma_1(1) = z_0$ where γ_1 is only located in the upper quadrants of the complex plane.

Similarly, $\gamma_2[0,1]$ is a curve where $\gamma_2(0) = 1$ and $\gamma_2(1) = z_0$ and $\gamma_2[0,1]$ is only located in the lower quadrants of the complex plane. We will now use the integral definition of log to study the discontinuity that occurs at any $z \in (-\infty, 0)$. Where:

$$\begin{aligned}
\lim_{t\to 1^-}\Big(\log(\gamma_1(t))\Big) - \lim_{t\to 1^-}\Big(\log(\gamma_2(t))\Big) &= \lim_{t\to 1^-}\left(\int_{\gamma_1[0,t]} \frac{1}{z}\right) - \lim_{t\to 1^-}\left(\int_{\gamma_2[0,t]} \frac{1}{z}\right) \\
&= \lim_{t\to 1^-}\left(\int_{\gamma_1[0,t]} \frac{1}{z}\right) - \lim_{t\to 1^-}\left(\int_{\gamma_2[0,t]} \frac{1}{z}\right) + 0 \\
\text{As } \frac{1}{z} \text{ is well defined at } z_0\text{:} & \\
&= \lim_{t\to 1^-}\left(\int_{\gamma_1[0,t]} \frac{1}{z}\right) - \lim_{t\to 1^-}\left(\int_{\gamma_2[0,t]} \frac{1}{z}\right) + \left(\int_{z_0} \frac{1}{z} - \int_{z_0} \frac{1}{z}\right) \\
&= \left(\int_{\gamma_1} \frac{1}{z}\right) - \left(\int_{\gamma_2} \frac{1}{z}\right)
\end{aligned}$$

$\gamma_1[0,1] - \gamma_2[0,1]$ produces a closed curve encompassing 0. $\frac{1}{z}$ has one simple pole at $z = 0$ where $\lim_{z\to 0}(z-0)\frac{1}{z} = \lim_{z\to 0} 1 = 1$.

$$\begin{aligned}
\int_{\gamma_1} \frac{1}{z} - \int_{\gamma_2} \frac{1}{z} &= \int_{\gamma_1-\gamma_2} \frac{1}{z} \\
&= 2\pi i(Res(0; \frac{1}{z})) \\
&= 2\pi i && \text{By residue theorem}
\end{aligned}$$

Thus, we have shown that $\lim_{t\to 1^-}\Big(\log(\gamma_1(t))\Big) - \lim_{t\to 1^-}\Big(\log(\gamma_2(t))\Big) = 2\pi i$; in other words, there is a $2\pi i$ jump discontinuity at z_0. However, this jump discontinuity can be removed by subtracting $i\arg(z)$. Which brings us to our next claim, $\log(z) - i\arg_{(-\pi,\pi)}(z)$ is continuous everywhere except 0.

Proof. Firstly, as it is already known that $-i\arg_{(-\pi,\pi)}(z)$ and $\log(z)$ continuous on $\mathbb{C}\backslash(-\infty, 0]$; thus, $\log(z) - i\arg_{(-\pi,\pi)}(z)$ is continuous on $\mathbb{C}\backslash(-\infty, 0]$. Now, given any $z_0 \in (-\infty, 0)$ chose any 2 curves that posses the same

properties as the aforementioned $\gamma_1[0,1]$ and $\gamma_2[0,1]$ where:

$$\lim_{t\to 1^-}\Big(\log(\gamma_1(t)) - i\arg_{(-\pi,\pi)}(\gamma_1(t))\Big) - \lim_{t\to 1^-}\Big(\log(\gamma_2(t)) - i\arg_{(-\pi,\pi)}(\gamma_2(t))\Big) =$$
$$= \lim_{t\to 1^-}\Big(\log(\gamma_1(t))\Big) - \lim_{t\to 1^-}\Big(i\arg_{(-\pi,\pi)}(\gamma_1(t))\Big) - \lim_{t\to 1^-}\Big(\log(\gamma_2(t))\Big) + \lim_{t\to 1^-}\Big(i\arg_{(-\pi,\pi)}(\gamma_2(t))\Big)$$
$$= 2\pi i - \lim_{t\to 1^-}\Big(i\arg_{(-\pi,\pi)}(\gamma_1(t))\Big) + \lim_{t\to 1^-}\Big(i\arg_{(-\pi,\pi)}(\gamma_2(t))\Big)$$
$$= 2\pi i - 2\pi i + 0$$
$$= 0$$

Thus, as γ_1 is an arbitrary curve approaching z_0 from above and γ_2 is an arbitrary curve approaching z_0 from below, it is shown that. $\lim_{z\to z_0}\log(z) - i\arg_{(-\pi,\pi)}(z)$. Thus, it is shown that $\log(z) - i\arg_{(-\pi,\pi)}(z)$ is continuous on $\mathbb{C}\backslash\{0\}$. $\square$

A similar framework can be applied to the dilogarithm to produce Bloch-Wigner functions. As we have previously stated that the dilogarithm is analytic on $\mathbb{C}\backslash(1,\infty)$. Using the integral definition of dilogarithm we shall consider looking at the difference of the dilogarithm path integral approaching the branch cut from below, and above to converge at $z_0 \in (1,\infty)$. We decided to use the smooth piece-wise curve approaching from below as $\gamma : [-1,1] \to \mathbb{C}\backslash(1,-\infty)$ where $\gamma[-1,1] = \gamma_1[-1,0] + \gamma_2[0,z_0]$ such that $\gamma_1(t) = -i\varepsilon(1+t)$ and $\gamma_2(t) = t - \varepsilon i$. Then we defined a smooth piece wise curve approaching from above as $\gamma^* : \mathbb{R} \to \mathbb{C}\backslash(1,\infty)$ where $\gamma_1^*[-1,0] + \gamma_2^*[0,z_0]$ such that $\gamma_1^*(t) = -i\varepsilon(1+t)$ and $\gamma_2^*(t) = t - i\varepsilon$ where the curves are illustrated in 1 :

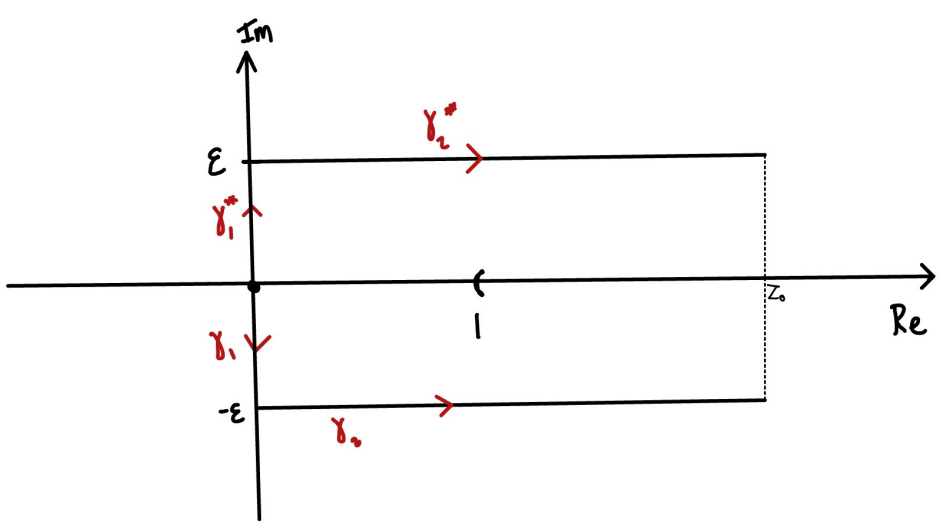

Figure 1: Depiction of γ and γ^*

$$\lim_{\varepsilon\to 0}\left(\int_{\gamma}\frac{-\log(1-z)}{z}dz-\int_{\gamma^*}\frac{-\log(1-z)}{z}dz\right)$$
$$=\lim_{\varepsilon\to 0}\left(\int_{\gamma_1}\frac{-\log(1-z)}{z}dz+\int_{\gamma_2}\frac{-\log(1-z)}{z}dz-\int_{\gamma_1^*}\frac{-\log(1-z)}{z}dz-\int_{\gamma_2^*}\frac{-\log(1-z)}{z}dz\right)$$
$$=\lim_{\varepsilon\to 0}\left(\int_{-1}^{0}\frac{-\log(1+i\varepsilon(t+1))}{(t+1)}dz-\int_{-1}^{0}\frac{-\log(1-i\varepsilon(t+1))}{(t+1)}dz\right)+\lim_{\varepsilon\to 0}\left(\int_{\gamma_2}\frac{-\log(1-z)}{z}dz-\int_{\gamma_2^*}\frac{-\log(1-z)}{z}dz\right)$$

By dominated convergence theorem:

$$=\int_{-1}^{0}\frac{\lim_{\varepsilon\to 0}\Big(-\log(1+i\varepsilon(t+1))+\log(1-i\varepsilon(t+1))\Big)}{(t+1)}dz+\lim_{\varepsilon\to 0}\left(\int_{\gamma_2}\frac{-\log(1-z)}{z}dz-\int_{\gamma_2^*}\frac{-\log(1-z)}{z}dz\right)$$

As $\log(1-z)$ is continuous at $z=0$:

$$=\int_{-1}^{0}\frac{0}{(t+1)}dz+\lim_{\varepsilon\to 0}\left(\int_{\gamma_2}\frac{-\log(1-z)}{z}dz-\int_{\gamma_2^*}\frac{-\log(1-z)}{z}dz\right)$$
$$=\lim_{\varepsilon\to 0}\left(\int_{0}^{z_0}\frac{-\log(1-(t-\varepsilon i))}{t-\varepsilon i}dt-\int_{0}^{z_0}\frac{-\log(1-(t+\varepsilon i))}{t+\varepsilon i}dt\right)$$
$$=\lim_{\varepsilon\to 0}\left(\int_{0}^{z_0}\frac{-(t+\varepsilon i)\log(1-(t-\varepsilon i))+(t-\varepsilon i)\log(1-(t+\varepsilon i))}{(t-\varepsilon i)(t+\varepsilon i)}dt\right)$$

By dominated convergence theorem:

$$=\int_{0}^{z_0}\lim_{\varepsilon\to 0}\left(\frac{-(t+\varepsilon i)\log(1-(t-\varepsilon i))+(t-\varepsilon i)\log(1-(t+\varepsilon i))}{(t-\varepsilon i)(t+\varepsilon i)}\right)dt$$
$$=\int_{0}^{z_0}\frac{-\lim_{\varepsilon\to 0}(t+\varepsilon i)\lim_{\varepsilon\to 0}\log(1-(t-\varepsilon i))+\lim_{\varepsilon\to 0}(t-\varepsilon i)\lim_{\varepsilon\to 0}\log(1-(t+\varepsilon i))}{\lim_{\varepsilon\to 0}(t-\varepsilon i)\lim_{\varepsilon\to 0}(t+\varepsilon i))}dt$$
$$=\int_{0}^{z_0}\frac{-t\lim_{\varepsilon\to 0}\log(1-(t-\varepsilon i))+t\lim_{\varepsilon\to 0}\log(1-(t+\varepsilon i))}{t^2}dt$$
$$=\int_{0}^{z_0}\frac{-\lim_{\varepsilon\to 0}\log(1-(t-\varepsilon i))+\lim_{\varepsilon\to 0}\log(1-(t+\varepsilon i))}{t}dt$$
$$=\int_{0}^{z_0}\frac{\lim_{\varepsilon\to 0}\big(\log|1-(t+\varepsilon i)|-\log|1-(t-\varepsilon i)|\big)+\lim_{\varepsilon\to 0}\big(\arg_{(-\pi,\pi)}(1-(t+\varepsilon i))-\arg_{(-\pi,\pi)}(1-(t-\varepsilon i))\big)}{t}dt$$

As $\log|z|$ has been shown to be continuous on $\mathbb{C}\backslash\{0\}$. Therefore, $\lim_{\varepsilon\to 0}\big(\log|1-(t+\varepsilon i)|-\log|1-(t-\varepsilon i)|\big)=0$.

$$=\int_{0}^{z_0}\frac{\lim_{\varepsilon\to 0}\big(\arg_{(-\pi,\pi)}(1-(t+\varepsilon i))-\arg_{(-\pi,\pi)}(1-(t-\varepsilon i))\big)}{t}dt$$

As we know that $\arg_{(-\pi,\pi)}$ has a jump of 2π when crossing through $(-\infty,0)$ so, $\arg_{(-\pi,\pi)}(1-z)$ has a jump of 2π when crossing $(1,\infty)$. Hence, as $\lim_{\varepsilon\to 0}\big(\arg_{(-\pi,\pi)}(1-(z+\varepsilon i))-\arg_{(-\pi,\pi)}(1-(z-\varepsilon i))\big)$ is the difference of the argument from opposing sides (from above and below) of $\arg_{(-\pi,\pi)}$'s -2π jump discontinuity; therefore, $\lim_{\varepsilon\to 0}\big(\arg_{(-\pi,\pi)}(1-(z+\varepsilon i))-\arg_{(-\pi,\pi)}(1-(z-\varepsilon i))\big)=-2\pi$. Hence:

$$=\int_{0}^{z_0}\left(\frac{-2\pi i}{t}\right)dt$$
$$=-2\pi i\log(z_0)$$
$$=-2\pi i\log|z_0| \qquad \text{As } z\in\mathbb{R} \text{ thus } |z_0|=z_0$$

Therefore, there is a jump discontinuity of $-2\pi i \log|z_0|$ as we approach the dilogarithms $(1,\infty)$ branch cut from above relative to when we approached it from below from 2 specific curves. Now, we will demonstrate that this jump discontinuity is the same independent of the curve selection to approach the discontinuity. We will, without loss of generality, prove this by showing that the path integral over $\frac{-\log(1-z)}{z}$ with the arbitrary curve approaching the branch cut from above will have have the same value path integral value as γ^* over $\frac{-\log(1-z)}{z}$ as ε approaches zero.

Define an arbitrary curve piecewise smooth curve ψ where for $a, b \in \mathbb{R}$ such that $a < b$ $\psi : [a,b) \to \mathbb{C}\backslash(1,\infty)$ where $\psi(a) = 0$, $\lim_{z^- \to b}\big(\psi(z)\big) = z_0$ and ψ approaches the branch cut from above the branch cut illustrated in 2.

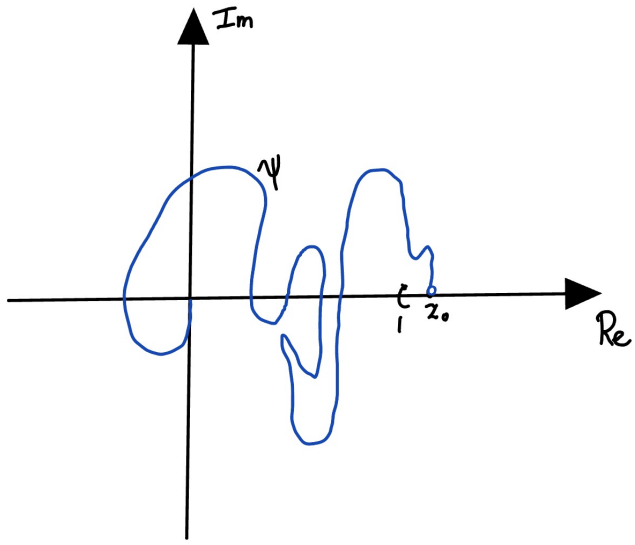

Figure 2: Depiction of ψ

Now, let $\varepsilon > 0$ and consider $D(z_0, 2\varepsilon)$ as depicted in 3:

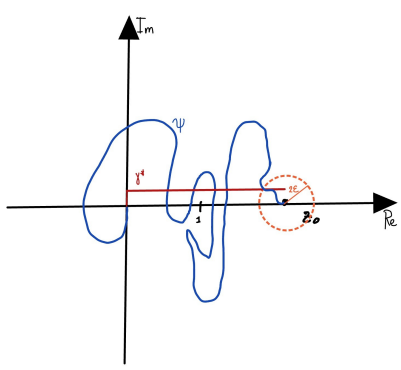

Figure 3:

Now, choose $\alpha \in [a,b)$ such that $\psi(\alpha) \in D(z_0; 2\varepsilon)$. Now define the line segment $l : [0,1] \to \mathbb{C}\backslash(1,\infty)$ where $l(0) = \psi(\alpha)$ and $l(1) = \gamma^*(z_0)$ where this can be seen 4:

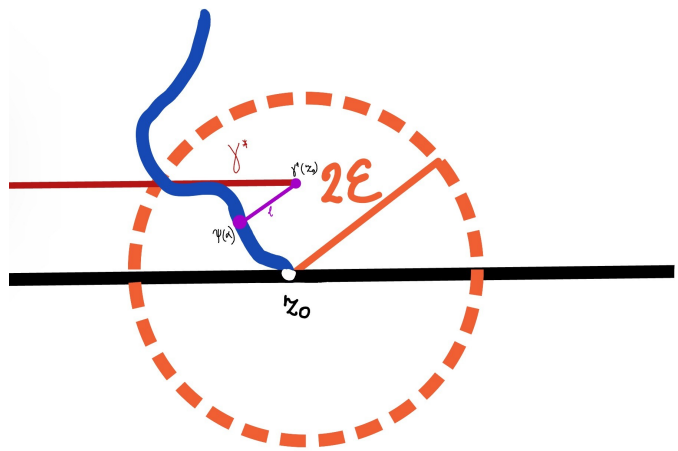

Figure 4:

Thus, $\psi[a,\alpha] + l[0,1] - \gamma^*[-1,z_0]$ produces a closed curve in $\mathbb{C}\backslash(1,\infty)$ where by Cauchy's theorem :

$$\oint_{\psi[a,\alpha]+l[0,1]-\gamma^*[-1,z_0]} \frac{-\log(1-z)}{z} dz = 0$$

Therefore the following can be deduced:

$$\oint_{\psi[a,\alpha]+l[0,1]-\gamma^*[-1,z_0]} \frac{-\log(1-z)}{z} dz = 0 \Leftrightarrow$$

$$\Leftrightarrow -\int_{\psi[a,\alpha]} \frac{-\log(1-z)}{z} dz + \int_{\gamma^*[-1,z_0]} \frac{\log(1-z)}{z} dz = \int_{l[0,1]} \frac{\log(1-z)}{z} dz$$

$$\Leftrightarrow \left| \int_{\psi[a,\alpha]} \frac{-\log(1-z)}{z} dz - \int_{\gamma^*[-1,z_0]} \frac{\log(1-z)}{z} dz \right| = \left| \int_{l[0,1]} \frac{\log(1-z)}{z} dz \right|$$

$$\Leftrightarrow \left| \int_{\psi[a,\alpha]} \frac{-\log(1-z)}{z} dz - \int_{\gamma^*[-1,z_0]} \frac{\log(1-z)}{z} dz \right| \leq \int_{l[0,1]} \left| \frac{\log(1-z)}{z} \right| dz$$

By Extreme value theorem there exists $M \in \mathbb{R}$ where $\left| \frac{-\log(1-z)}{z} \right| < M$ for all $z \in l[0,1]$

$$\Leftrightarrow \left| \int_{\psi[a,\alpha]} \frac{-\log(1-z)}{z} dz - \int_{\gamma^*[-1,z_0]} \frac{\log(1-z)}{z} dz \right| \leq M \cdot length(l) < 2M\varepsilon$$

Thus, when $\varepsilon > 0$ is shrunk to 0, the two integrals attain the same value. Thus, it has been proven the discontinuity would remain the same regardless of the choice of path to z_0. Therefore, we have established that there exists $2\pi i \log|z|$ jump discontinuity on the branch cut on $(1,\infty)$.

Thus, similarly to the log motivation, we will claim adding $i \arg_{(-\pi,\pi)}(1-z) \log|z|$ to the logarithm will resolve this jump discontinuity resulting $Li_2(z) + i \log|z| \arg_{(-\pi,\pi)}(1-z)$ being continuous on $\mathbb{C}\backslash\{0\}$.

Proof. Firstly, given that $Li_2(z)$, $i \log|z|$ and $\arg(1-z)$ are continuous on $\mathbb{C}\backslash(1,\infty) \cup \{0\}$; thus, $Li_2(2) + i \arg(1-z) \log|z|$ is continuous on $Li_2(z) + i \log|z| \arg_{(-\pi,\pi)}(z)$.

Now, when approaching the branch cut from below and above the jump discontinuity there is no longer a jump discontinuity for any $z_0 \in (1, \infty)$:

$$\lim_{z\to z_0, \Im(z)<0} \Big(Li_2(z) + i \arg_{(-\pi,\pi)}(1-z) \log|z| \Big) - \lim_{z\to z_0, \Im(z)<0} \Big(Li_2(z) + i \arg_{(-\pi,\pi)}(1-z) \log|z| \Big) =$$

$$= \lim_{z\to z_0, \Im(z)<0} Li_2(z) - \lim_{z\to z_0, \Im(z)>0} Li_2(z) + \lim_{z\to z_0, \Im(z)<0} \big(i \arg_{(-\pi,\pi)}(1-z) \log|z| \big) - \lim_{z\to z_0, \Im(z)>0} \big(i \arg_{(-\pi,\pi)}(1-z) \log|z| \big)$$

As there is a jump discontinuity of $-2i\pi \log|z_0|$ at the branch cut is $(1, \infty)$

$$= 2i\pi \log|z_0| + \lim_{z\to z_0, \Im(z)<0} \big(i \arg_{(-\pi,\pi)}(1-z) \log|z| \big) - \lim_{z\to z_0, \Im(z)>0} \big(i \arg_{(-\pi,\pi)}(1-z) \log|z| \big)$$

$$= -2i\pi \log|z_0| + i \log|z_0| \Bigg(\lim_{z\to z_0, \Im(z)<0} \big(\arg_{(-\pi,\pi)}(1-z) \big) - \lim_{z\to z_0, \Im(z)>0} \big(\arg_{(-\pi,\pi)}(1-z) \big) \Bigg)$$

$$= -2\pi i \log|z_0| + i \log|z_0| \lim_{z\to z_0, \Im(z)<0} \Big(\arg_{(-\pi,\pi)}(1-z) - 0 \Big)$$

$$= -2\pi i \log|z_0| + i \log|z_0| \lim_{z\to z_0, \Im(z)<0} \Big(\arg_{(-\pi,\pi)}(1-z) \Big)$$

$$= -2\pi i + 2\pi i$$

$$= 0$$

Thus, the jump over the branch cut has been resolved so we have proven that $Li_2(z) + i \arg(1-z) \log|z|$ is continuous on $\mathbb{C}\backslash\{0\}$. Thus, the imaginary component is continuous too; hence, $\Im(Li_2(z)) + i \arg(1-z) \log|z|$ is continuous on $\mathbb{C}\backslash\{0\}$. Furthermore, D satisfies similar equalities to $Li_2(z)$, but without log and constants terms making the equalities much simpler some of which we will now explore. □

Proposition 8.

$$D(\frac{1}{z}) = -D(z)$$

Proof.

$$\begin{aligned}
D(\frac{1}{z}) &= \Im\Big(Li_2(\frac{1}{z}) + iarg(1-\frac{1}{z}) \log\big(|\frac{1}{z}|\big) \Big) \\
&= \Im\Big(-Li_2(z) - \frac{\pi^2}{6} - \frac{1}{2}\log^2(1-z) + arg\Big(1-\frac{1}{z}\Big) \log\Big|\frac{1}{z}\Big| \Big) && \text{By 1.4.2} \\
&= \Im(-Li_2(z)) - \Im\big(\frac{1}{2}\log^2(1-z)\big) + arg(1-\frac{1}{z}) \log\big(|\frac{1}{z}|\big) \\
&= \Im(-Li_2(z)) - \Im\Big(\frac{1}{2}\Big(\log^2(|-z|) + 2i\log(|-z|)arg(-z) - arg(|-z|) \Big)\Big) + arg(1-\frac{1}{z}) \log\big(|\frac{1}{z}|\big) \\
&= \Im(-Li_2(z)) - \log(|-z|)arg(-z) + arg(1-\frac{1}{z}) \log\big(|\frac{1}{z}|\big) \\
&= -\Im(Li_2(z)) - \log(|-z|)arg(-z) - arg(\frac{z-1}{z}) \log(|-z|) \\
&= -\Im(Li_2(z)) - \log(|-z|)arg(-z) - arg(z-1)\log(|-z|) + arg(z)\log(|-z|) \\
&= -\Im(Li_2(z)) - arg(z-1)\log(|-z|) \\
&= -D(z)
\end{aligned}$$

□

$$D(1-z) = -D(z)$$

Proof.

$$\begin{aligned}
D(1-z) &= \Im(Li_2(1-z)) + arg(1-1+z)\log(|1-z|) \\
&= \Im\Big(Li_2(1-z)\Big) + arg(z)\log(|1-z|) \\
&= \Im\Big(\frac{\pi^2}{6} - \log(z)\log(1-z) - Li_2(z)\Big) + arg(z)\log(|1-z|) \\
&= -arg(z)\log(|1-z|) - arg(1-z)\log(|z|) - \Im\Big(Li_2(z)\Big) + arg(z)\log(|1-z|) \\
&= -\Im\big(Li_2(z)\big) - arg(1-z)\log(|z|) \\
&= -D(z)
\end{aligned}$$

□

Now we can generalize the the duplication formula proven for $Li_k(z)$.

Proposition 9 (Distribution Formula)**.** $D(x^n) = n\sum_{k=0}^{n-1} D(e^{\frac{2\pi k i}{n}}x)$

Proof. Firstly, we will prove that $Li_2(x^n) = n\sum_{k=0}^{n-1} Li_2(e^{\frac{2\pi k i}{n}}x)$.

$$\begin{aligned}
(Li_2(x^n))' &= \frac{nx^{n-1}}{x^n} Li_1(x^n) && \text{By chain rule} \\
&= -\frac{n}{x}\ln(1-x^n) \\
&= -\frac{n}{x}\ln\big(\prod_{k=0}^{n-1}(e^{\frac{k2\pi i}{n}} - x)\big) \\
&= -\frac{n}{x}\big(\sum_{k=0}^{n-1}\ln(e^{\frac{k2\pi i}{n}} - x)\big) \\
&= -\frac{n}{x}\Bigg(\sum_{k=0}^{n-1}\ln\big(e^{\frac{k2\pi i}{n}}(1-e^{-\frac{k2\pi i}{n}}x)\big)\Bigg) \\
&= -\frac{n}{x}\Bigg(\sum_{k=0}^{n-1}\big(\ln\big(e^{\frac{k2\pi i}{n}}\big) + \ln(1-e^{-\frac{k2\pi i}{n}}x)\big)\Bigg) \\
&= -\frac{n}{x}\Bigg(\sum_{k=0}^{n-1}\ln\big(e^{\frac{k2\pi i}{n}}\big) + \sum_{k=0}^{n-1}\ln(1-e^{-\frac{k2\pi i}{n}}x)\Bigg) \\
&= -\frac{n}{x}\Bigg(\ln\big(1\big) + \sum_{k=0}^{n-1}\ln(1-e^{-\frac{k2\pi i}{n}}x)\Bigg) \\
&= -\frac{n}{x}\Bigg(\sum_{k=0}^{n-1}\ln(1-e^{-\frac{k2\pi i}{n}}x)\Bigg)
\end{aligned}$$

Thus, by properties of Li_2:

$$\begin{aligned}
Li_2(x^n) &= -n\int_0^x \left(\frac{1}{t}\sum_{k=0}^{n-1}\ln(1-e^{-\frac{2k\pi i}{n}}t)\right)dt \\
&= -n\sum_{k=0}^{n-1}\int_0^x \left(\frac{1}{t}\ln(1-e^{-\frac{2k\pi i}{n}}t)\right)dt \\
&= -n\sum_{k=0}^{n-1}\int_0^{xe^{\frac{2k\pi i}{n}}} \left(\frac{1}{u}\ln(1-u)\right)du && \text{by setting } e^{\frac{2k\pi i}{n}}u = t \\
&= n\sum_{k=0}^{n-1}\int_0^{xe^{\frac{2k\pi i}{n}}} \Big(Li_1(u)\Big)du \\
&= n\sum_{k=0}^{n-1} Li_2(e^{\frac{2k\pi i}{n}}x)
\end{aligned}$$

Thus as $Li_2(x^n) = n\sum_{k=0}^{n-1} Li_2(e^{\frac{2k\pi i}{n}}x)$ then the imaginary components should agree as well thus:

$$\Im(Li_2(z^n)) = n\sum_{k=0}^{n-1}\big(Li_2(e^{\frac{2k\pi i}{n}}z)\big)$$

Now:

$$\begin{aligned}
i\arg(1-z^n)\log|z^n| &= i\arg\left(\prod_{k=0}^{n-1}(e^{\frac{2k\pi i}{n}}-z)\right)\log|z^n| \\
&= in\sum_{k=0}^{n-1}\arg\left(e^{\frac{2k\pi i}{n}}-z\right)\log|z| \\
&= in\sum_{k=0}^{n-1}\arg(e^{\frac{2k\pi i}{n}}(1-e^{-\frac{2\pi ik}{n}}))\log|z| \\
&= in\sum_{k=0}^{n-1}\left(\arg(e^{\frac{2k\pi i}{n}})\log|z| + \arg(1-e^{-\frac{2\pi ik}{n}})\log|z|\right) \\
&= in\sum_{k=0}^{n-1}\arg(e^{\frac{2k\pi i}{n}})\log|z| + in\sum_{k=0}^{n-1}\arg(1-e^{-\frac{2\pi ik}{n}})\log|z| \\
&= in\arg(\prod_{k=0}^{n-1}(e^{\frac{2k\pi i}{n}}))\log|z| + in\sum_{k=0}^{n-1}\arg(1-e^{-\frac{2\pi ik}{n}})\log|z| \\
&= in\arg(1)\log|z| + in\sum_{k=0}^{n-1}\arg(1-e^{-\frac{2\pi ik}{n}})\log|z| \\
&= in\sum_{k=0}^{n-1}\arg(1-e^{-\frac{2\pi ik}{n}})\log|z| \\
&= in\sum_{k=0}^{n-1}\arg(1-e^{\frac{2\pi ik}{n}})\log|z| \\
&= in\sum_{k=0}^{n-1}\left(\arg(1-e^{\frac{2\pi ik}{n}})\log|e^{2k\pi i}nz|\right) && \text{As } |e^{\frac{2\pi ik}{n}}| = 1 \text{ so } |z| = |z||e^{\frac{2\pi ik}{n}}| = |ze^{\frac{2\pi ik}{n}}|
\end{aligned}$$

Thus, we have shown that:

$$\Im(Li_2(z^n)) + i\arg(1-z^n)\log|z| = n\sum_{k=0}^{n-1}\left(\Im(Li_2(e^{\frac{2k\pi i}{n}})) + i\arg(1-e^{\frac{2k\pi i}{n}})\log|e^{\frac{2k\pi i}{n}}|\right) \Leftrightarrow D(x^n) = n\sum_{k=0}^{n-1} D(e^{\frac{2k\pi i}{n}}x)$$

. □

5 Part Four: Abstraction of the Dilogarithm

So far, we have proven properties of D using complex analysis, however, in fact all these properties can be derived directly from the Five-term relation. Using this will allow us to abstract the di-logarithm to any field.

Proposition 10 (Six Fold Symmetry)**.** *To begin we shall show that we are able to derive the six fold symmetry of the Bloch-Wigner function which is for $z \in \mathbb{C}\backslash\{0,1\}$:*

$$D(z) = D(1-\frac{1}{z}) = D(\frac{1}{1-z}) = -D(\frac{1}{z}) = -D(1-z) = -D(\frac{-z}{1-z})$$

Holds (We use the fact that $D(0) = D(\infty) = D(1) = 0$, as shown in [1]).

By substituting $x = z$ and $y = 0$ into the Five-term relation we can establish that:

$$\begin{aligned} D(z) + D(0) + D(\frac{1-z}{1-0}) + D(\frac{1-0}{1}) + D(1) = 0 &\Leftrightarrow D(z) = -D(1-z) - 2D(1) - D(0) \\ &\Leftrightarrow D(z) = -D(1-z) \end{aligned}$$

Similarly, by substituting $x = \frac{1}{z}$ and $y = z$ into the Five-term Relation we deduce that:

$$\begin{aligned} D(\frac{1}{z}) + D(z) + D(1-\frac{1}{z}(z)) + D(\frac{1-\frac{1}{z}}{1-\frac{1}{z}(z)}) + D(\frac{1-z}{1-\frac{1}{z}(z)}) = 0 &\Leftrightarrow D(\frac{1}{z}) + D(z) = 0 \\ &\Leftrightarrow D(\frac{1}{z}) = -D(z) \end{aligned}$$

Now we can use these equalities to establish that:

$$\begin{aligned} D(1-\frac{1}{z}) &= -D(\frac{1}{z}) \\ &= (-1)(-1)D(z) \\ &= D(z) \end{aligned}$$

$$\begin{aligned} D(\frac{1}{1-z}) &= -D(1-z) \\ &= (-1)(-1)D(z) \\ &= D(z) \end{aligned}$$

$$
\begin{aligned}
-D(\frac{-z}{1-z}) &= D(1+\frac{z}{z+1}) \\
&= D(\frac{1-z+z}{1-z}) \\
&= D(\frac{1}{1-z}) \\
&= D(z)
\end{aligned}
$$

Thus we have successfully derived the Six Fold Symmetry only using the Five-term Relation. Another example we explored was deriving z^2 from the Five-term relation. We start by setting $x, y = z$ in the Five-term relation resulting in the following:

$$D(z) + D(z) + D(1-(z)(z)) + D(\frac{1-z}{1-(z)(z)}) + D(\frac{1-z}{1-(z)(z)}) = 0$$

$$\Leftrightarrow 2D(z) + D(1-z^2) + 2D((\frac{1}{z+1}) = 0$$

$$\Leftrightarrow 2D(z) + D(1-z^2) + 2D(-z) = 0 \qquad \text{By Six-Fold Symmetry } D(\frac{1}{1+(-z)}) = D(-z)$$

$$\Leftrightarrow 2D(z) + 2D(-z) = -D(1-z^2)$$

$$\Leftrightarrow 2D(z) + 2D(-z) = D(z^2) \qquad \text{By Six-Fold Symmetry}$$

This example hints at the fact that the Five-term Relation allows us to describe rational functions in terms of simpler expressions of D . Before we discuss this further, we will have to analyse the cross ratio and its relevance to to Bloch-Winger functions.

A Möbius, transformation is $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$ is a rational function.

$$f(z) = \frac{az+b}{cz+d} \qquad a, b, c, d \in \mathbb{C}$$

Where f is a bijection. Now, every Möbius transformation can be depicted in $PLG(2, \mathbb{C})$ as follows:

$$f(z) = \frac{a_f z + b_f}{c_f z + d_f}$$

Can be represented as:

$$\begin{bmatrix} a_f & b_f \\ c_f & d_f \end{bmatrix}$$

Proposition 11. *Where, given g, f be Möbius transformations, denoted by Matrices: A_f, A_g. Then, $f \circ g$ has $A_f \times A_g$ as the matrix representation and that composition of Möbius transformations are Möbius.*

Proof. Define two Möbius transformations $f, g : \mathbb{C}_\infty \to \mathbb{C}_\infty$. Where $f(z) = \frac{a_f z + b_f}{c_f z + d_f}$ and $g(z) = \frac{a_g z + b_g}{c_g z + d_g}$; hence:

$$\begin{aligned} f \circ g(z) &= \frac{a_f(\frac{a_g(z)+b_z}{c_g(z)+d_g}) + b_f}{c_f(\frac{a_g(z)+b_z}{c_g(z)+d_g}) + d_f} \\ &= \frac{(a_f a_g + b_f c_g)z + (a_f b_g + b_f d_g)}{(c_f a_g + c_g d_f)z + (c_f b_g + d_f d_g)} \end{aligned}$$

Which has the matrix representation: $\begin{bmatrix} (a_f a_g + b_f c_g) & (a_f b_g + b_f d_g) \\ (c_f a_g + c_g d_f) & (c_f b_g + d_f d_g) \end{bmatrix} = \begin{bmatrix} a_f & b_f \\ c_f & d_f \end{bmatrix} \times \begin{bmatrix} a_g & b_g \\ c_g & d_g \end{bmatrix}$

□

Proposition 12. *When considering a Möbius transformation $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$ then given the distinct points: $z_1, z_2, z_3 \in \mathbb{C}_\infty$ and set:*

- $f(z_1) = 1$
- $f(z_2) = 0$
- $f(z_3) = \infty$

Then, only one Möbius transformation will satisfy these equalities.

Proof. Firstly, the existence of such a Möbius transformation will be shown:

Given that $f(z_2) = 0$ so $f(z) = z - z_2$. Next as $f(z_3) = \infty$ as well then $f(z) = \frac{z - z_1}{z - z_3}$. Additionally, as $f(z_1) = 1$ then:

$$f(z) = \frac{z - z_2}{z - z_3} \times \frac{z_1 - z_3}{z_1 - z_2} = \frac{(z_1 - z_3)(z) - z_2(z_1 - z_3)}{(z_1 - z_2)(z) - z_3(z_1 - z_2)}$$

Secondly, the uniqueness of such a transformation will be proven. Suppose, for a contradiction, there are two distinct Möbius transformations that satisfy the the following:

- $f(z_1) = 1 = g(z_1)$
- $f(z_2) = 0 = g(z_2)$
- $f(z_3) = \infty = g(z_3)$

As g^{-1} and f are Möbius transformations thus, $g^{-1} \circ f(z)$ is a Möbius transformation as well (non identity as $f \neq g$). Thus, there exists $a, b, c, d \in \mathbb{C}$ (where $ac - bd \neq 0$) such that:

$$\begin{aligned} \frac{az_i + b}{cz_i + d} = z_i &\Leftrightarrow az_i + b = z_i(cz_i + d) \qquad\qquad i \in \{1, 2, 3\} \\ &\Leftrightarrow az_i + b = cz_i^2 + dz_i \\ &\Leftrightarrow 0 = cz_i^2 + (d - a)z_i - b \end{aligned}$$

By Fundamental Theorem of Algebra, if $g^{-1} \circ f$ isn't an identity, mapping it $g^{-1} \circ f$ will have at most 2 fixed points. However, $g^{-1} \circ f$ has three distinct points (z_1, z_2, z_3), forcing the function to be the identity mapping. Thus, by the definition of inverses, this forces $g = f$, which is a contradiction. Thus, the Möbius transformation must be unique. $\square$

Definition 2 (Cross Ratio)**.** *The cross ratio of four distinct ordered points in the Riemann sphere, (z_1, z_2, z_3, z_4) is the element $z \in \mathbb{C}\backslash(0,1)$ such that there is a Möbius transformation f with fixed mapping to four points $(z_1, z_2, z_3, z_4 \in \mathbb{C}_\infty)$:*

- $f(z_0) = z$
- $f(z_1) = 1$
- $f(z_2) = \infty$
- $f(z_3) = 0$

Where by applying the method as when fixing three and combining this with $f(z_0) = z$ the following can be obtained:

$$f(z_0) = \frac{(z_1 - z_3)(z_0) - z_2(z_1 - z_3)}{(z_1 - z_2)(z_0) - z_3(z_1 - z_2)} = z \Leftrightarrow \frac{(z_0 - z_2)}{(z_0 - z_3)} \frac{(z_1 - z_3)}{(z_1 - z_2)} = z$$

Definition 3. $\tilde{D} : Conf_4 \to \mathbb{R}$ *such that* $\tilde{D}(z_0, z_1, z_2, z_3) = D\left(\frac{(z_3 - z_2)}{(z_3 - z_1)} \frac{(z_4 - z_1)}{(z_4 - z_2)}\right)$

Where we will use the cross ratio to lift the definition of $\tilde{D}$ from $\mathbb{C}\backslash\{0,1\}$ to the space of configurations of four points in the Riemann sphere. Now that all these terms have been established, we can show the a remarkable proposition using the Five-term relation:

Proposition 13. *For $f \in \mathbb{C}_\infty(t)$ where n is the order of f: $D(f(t)) = \frac{1}{n} \sum_{\substack{z_1 \in f^{-1}(1) \\ z_2 \in f^{-1}(0) \\ z_3 \in f^{-1}(\infty)}} \widetilde{D}(z, z_1, z_2, z_3)$ (Note: terms of multiplicity m in the preimage are considered separate elements in the sum)*

Where we can demonstrate this proposition using $D(z^2)$. $f(z) = z^2$ is a function of order 2 and that:

- $f^{-1}(1) = \{1, -1\}$
- $f^{-1}(0) = \{\infty\}$ with multiplicity of 2

- $f^{-1}(\infty) = \{0\}$ with multiplicity of 2

$$
\begin{aligned}
D(f(z)) &= \frac{1}{2} \sum_{\substack{z_1 \in \{1,-1\} \\ z_2 \in \{\infty\} \\ z_3 \in \{0\}}} \widetilde{D}(z, z_1, z_2, z_3) \\
&= \frac{1}{2} \sum_{z_1 \in \{1,-1\}} 2(2)\widetilde{D}(z, z_1, \infty, 0) \\
&= \frac{1}{2}\Big(4\widetilde{D}(z, 1, \infty, 0) + (2)(2)\widetilde{D}(z, -1, \infty, 0)\Big) \\
&= \Big(2\widetilde{D}(z, 1, \infty, 0) + (2)\widetilde{D}(z, -1, \infty, 0)\Big)
\end{aligned}
$$

Now we will prove the proposition:

Proof. Define $f(t) \in \mathbb{C}(t)$ where $[f] \in \mathbb{Z}[\mathbb{C}(t)]/(\text{Five-term Relation})$. Thus, there exists complex polynomials $A(t), B(t)$ where both are polynomials of degree less than or equal to n and where at least one of the two are non-constant and $A(t)$ is the numerator of f and $B(t)$ is the denominator of f. Now, Define $C(t) := B(t) - A(t)$. Where we claim that $C(t)$ and $A(t)$ are nonconstant which results in two cases.

Given the case that $A(t)$ is constant then by definition f that means that $B(t)$ is non-constant.

Thus as $D(f(z)) = -D(\frac{1}{f(z)})$ by Five-term relation, thus $[f] \equiv -[\frac{1}{f}]$ (by Six-fold symmetry) where without loss generality replace f with $\frac{1}{f}$. In the case that $C(t)$ may be constant with our initial choice of f. This would happen if $B(t)$, $A(t)$ agree up to a constant (thus neither are constant). As a result, $\frac{1}{1-f(z)} = \frac{B(t)}{1-A(t)}$ where $B(t)-(1-A(t)) = B(t)+A(t)-1$ will not be a constant. As the Six-fold symmetry states that $D(\frac{1}{1-f(z)}) = D(f(z))$, $[\frac{1}{1-f(z)}] \equiv [f(z)]$. Thus, without loss of genrality replace f with $\frac{1}{1-f(z)}$.

Now, given that $A(t)$ and $C(t)$ are non-constant, by fundamental theorem of algebra let $a, c \in \mathbb{C}$ where a is a root of $A(t)$ and c is a root $C(t)$. Now define $g(t) := \frac{c-a}{t-a}$, $A^*(t) := g(t)A(t)$ and $D(t) := B(t) - A^*(t)$.

$$
\begin{aligned}
A^*(t) &= \frac{c-a}{t-a}A(t) \\
&= \frac{c-a}{t-a}\Big((t-a)\phi(t)\Big) \qquad \text{where can be factored as } a \text{ is a root, } \phi \text{ is a polynomial of degree ¡ } n-1 \\
&= (c-a)\phi(t)
\end{aligned}
$$

Thus, $deg(A^*(t)) \leq n-1$. $deg(D(t)) \leq n$ as $deg(B(t)) \leq n$ and $A^*(t) \leq n-1$ and finally:

$$\begin{aligned} D(c) &= B(c) - A^*(c) \\ &= B(c) - g(c)A(C) \\ &= B(c) - \frac{c-a}{c-a}A(c) \\ &= C(c) \\ &= 0 && \text{As } c \text{ is root of } C \end{aligned}$$

Therefore, by plugging $x = f$ and $y = g$ into the Five-term relation we get that:

$$\begin{aligned} [f] + [g] + [1-fg] - [\frac{1-f}{1-fg}] - [\frac{1-g}{1-gf}] \equiv 0 \Leftrightarrow [f] &\equiv -[g] - [1-fg] - [\frac{1-f}{1-fg}] - [\frac{1-g}{1-gf}] \\ &\equiv -[g] + [fg] - [\frac{1-f}{1-fg}] + [1 - \frac{1-g}{1-gf}] && \text{by six-fold symmetry} \end{aligned}$$

From here by [3] we know that all the terms have degree of $n-1$ except potentially, the fg where $B(t)$ may be a polynomial of degree n. In that case we would just repeat the same proceeds for f by on fg after using the fact that $-[\frac{1}{fg}] \equiv [fg]$. Then we would substitute the $[fg]$ with the expression we derive. From here as we know that all the terms have a degree of $n-1$ or less that means we can apply our inductive hypothesis and define all the terms in terms of sums of $\tilde{D}$. From here we would use as series of computations to derive that $D(f(z)) = \frac{1}{n} \sum_{\substack{z_1 \in f^{-1}(1) \\ z_2 \in f^{-1}(0) \\ z_3 \in f^{-1}(\infty)}} \widetilde{D}(z, z_1, z_2, z_3)$. □

6 Conclusion

In conclusion, this research project has delved deeply into the fascinating world of the dilogarithm function, proving propositions from Don Zagier's work. By exploring the analytic continuation of polylogarithms and examining the unique properties of the dilogarithm, we have uncovered interesting insights into its behavior. The introduction of the Bloch-Wigner function has further expanded our understanding, offering a continuous extension over a broader domain with simpler equalities. The five-term relation has proven to be a powerful tool, allowing us to derive properties of the dilogarithm without it needing to be exclusively defined on the complex plane.

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